Practical 10

Aim: Consider a disk queue with requests for I/O to blocks on cylinders 98, 183, 41, 122, 14, 124, 65, 67. The SSTF scheduling algorithm is used. The head is initially at cylinder number 53. The cylinders are numbered from 0 to 199. Write a Program to calculate the total head movement (in number of cylinders) incurred while servicing these requests.

Code:
```
#include <stdio.h>
#include <stdlib.h>

int main() {
    int requests[] = {98, 183, 41, 122, 14, 124, 65, 67};
    int n = 8;
    int head = 53;
    int total_movement = 0;

    int visited[8] = {0};

    printf("Order of servicing:\n");

    for (int i = 0; i < n; i++) {
        int min_distance = 1000;
        int index = -1;

        // Find closest request
        for (int j = 0; j < n; j++) {
            if (!visited[j]) {
```

```
            int distance = abs(head - requests[j]);
            if (distance < min_distance) {
                min_distance = distance;
                index = j;
            }
        }
    }

    // Mark visited
    visited[index] = 1;

    printf("%d -> ", requests[index]);

    // Add movement
    total_movement += abs(head - requests[index]);

    // Move head
    head = requests[index];
  }

  printf("\n\nTotal Head Movement = %d\n", total_movement);

  return 0;
}
```

Output:

```
Order of servicing:
41 -> 65 -> 67 -> 98 -> 122 -> 124
    -> 14 ->

Total Head Movement = 323



=== Code Execution Successful ===
```